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Damage Mechanism Review - DEA Strength Increase, ATU-100

Scope and Objectives

This report documents the damage mechanism review conducted for ATU-100 in response to QSCP-MOC-2014-017. The objective is to evaluate the impact of increasing Diethanolamine (DEA) concentration from 25 to 30 wt% on the integrity of the rich and lean amine circuits. Robert Hendricks, as Unit Engineer, is responsible for defining the process parameters and identifying affected circuits, while Douglas Ferreira, as Inspection Supervisor, oversees the implementation of the revised inspection plan.

The review focuses on the potential for accelerated general corrosion and localized thinning due to higher acid-gas loading, as well as the risk of environmental cracking in non-stress-relieved piping. The assessment utilizes historical data from HIS-2011-0207 and technical guidance from API RP 571 to adjust the monitoring strategy for the Hamilton Plant.

Rich Amine Circuit Assessment

Lines 4-RA-101, 4-RA-102, and the header 6-RA-103 are expected to experience increased corrosion rates due to the higher acid-gas loading associated with QSCP-MOC-2014-017. Historical data from HIS-2011-0207 indicated that the rich amine circuit was already at half its remaining life. Consequently, the inspection interval for these lines is reduced to a 3-year thickness survey to comply with site rules for rich amine service.

Additional monitoring is required for 4-RA-104 at the L/R exchanger rich side. New CMLs will be established downstream of the let-down valve to monitor for erosion-corrosion and localized thinning. To provide real-time data on the increased corrosivity of the 30 wt% DEA solution, corrosion coupons will be installed on the rich-amine return header.

Lean Amine Cracking Screening

The lean amine circuit, specifically 4-LA-104, was screened for amine stress corrosion cracking as per API RP 571. Although the concentration change is the primary driver for the MOC, the screening for cracking is based on the operating temperature and the absence of post-weld stress relief. 4-LA-104 operates above 60 °C, which exceeds the threshold for cracking in non-stress-relieved carbon steel piping.

Douglas Ferreira has mandated external shear-wave UT for all circumferential and attachment welds on 4-LA-104. This NDE method is selected to detect internal cracking that surface methods like PT cannot characterize. The screening is independent of the DEA concentration increase and focuses on the susceptibility of the existing A106 Gr.B material under current thermal conditions.

Inspection Strategy and Responsibilities

Robert Hendricks shall ensure that the process conditions remain within the limits defined in QSCP-MOC-2014-017. Any deviation in acid-gas loading or temperature must be reported to the inspection department for further review. The retirement thickness for all lines, including 4-RA-101 and 4-RA-104, remains the t-min specified in the line facts; any reading below this value requires an immediate NCR.

Douglas Ferreira is responsible for updating the inspection database to reflect the 3-year interval for the rich amine lines. The next thickness survey will calculate the short-term corrosion rate using the data from HIS-2011-0207 as the baseline. All UT thickness measurements must be performed according to site rules to ensure accurate remaining life calculations and timely retirement of assets.

Sign-off	Name	Function	Date	Signature
Prepared	Robert Hendricks	Unit Engineer ATU-100, P.Eng.	2014-08-29	_____
Prepared	Douglas Ferreira	Inspection Supervisor, API 570 inspector	2014-08-29	_____